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Batch NO. 15

HDOC DAY-10 ACTIVITY REPORT: BATCHES 15 & 16

PART 1: TIME CONVERTER PROGRAM

1.1 Algorithm

1. Start

2. **Input:** Read total seconds (`total_seconds`).
3. **Calculate Days:** `days = total_seconds / 86400`
4. **Calculate Remaining Seconds:** `remaining_seconds = total_seconds % 86400`
5. **Calculate Hours:** `hours = remaining_seconds / 3600`
6. **Update Remaining Seconds:** `remaining_seconds = remaining_seconds % 3600`
7. **Calculate Minutes:** `minutes = remaining_seconds / 60`
8. **Calculate Seconds:** `seconds = remaining_seconds % 60`
9. **Output:** Display days, hours, minutes, and seconds.

10. Stop

1.2 Pseudocode

text

BEGIN

```
NUMBER total_seconds, days, hours, minutes, seconds  
OUTPUT "Enter time in seconds: "
```

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```
    INPUT total_seconds

    days = total_seconds / 86400
    hours = (total_seconds % 86400) / 3600
    minutes = ((total_seconds % 86400) % 3600) / 60
    seconds = ((total_seconds % 86400) % 3600) % 60

    OUTPUT days, "Days,", hours, "Hours,", minutes,
"Minutes,", seconds, "Seconds"
END
```

Use code with caution.

1.3 Flowchart Text Representation

text

```
[ Start ]
  |
  ▼
[ Input total_seconds ]
  |
  ▼
[ days = total_seconds / 86400 ]
[ rem_seconds = total_seconds % 86400 ]
  |
  ▼
[ hours = rem_seconds / 3600 ]
[ rem_seconds = rem_seconds % 3600 ]
  |
  ▼
[ minutes = rem_seconds / 60 ]
[ seconds = rem_seconds % 60 ]
  |
  ▼
[ Print Days, Hours, Minutes, Seconds ]
  |
  ▼
[ Stop ]
```

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Use code with caution.

1.4 Test Case Table

Test Case ID	Input	Expected Output	Status
TC1_01	0	0 Days, 0 Hours, 0 Minutes, 0 Seconds	Pass
TC1_02	45	0 Days, 0 Hours, 0 Minutes, 45	Pass
TC1_03	3750	0 Days, 1 Hours, 2 Minutes, 30	Pass
TC1_04	90061	1 Days, 1 Hours, 1 Minutes, 1 Seconds	Pass

1.5 Source Code (`time_converter.c`)

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The image shows a C program in a code editor and its execution output in a terminal. The program, named `day18.c`, takes a total number of seconds as input and converts it into days, hours, minutes, and seconds. The code uses integer division and modulo operations to calculate these values. The terminal output shows the program being compiled with `clang` and then executed multiple times with different inputs, demonstrating the conversion logic.

```
#include <stdio.h>

int main() {
    long long total_seconds;
    int days, hours, minutes, seconds;

    printf("Enter time in seconds: ");
    if (scanf("%lld", &total_seconds) != 1 || total_seconds < 0) {
        printf("Invalid input. Please enter a positive integer.\n");
        return 1;
    }

    days = total_seconds / 86400;
    int rem_seconds = total_seconds % 86400;
    hours = rem_seconds / 3600;
    rem_seconds = rem_seconds % 3600;
    minutes = rem_seconds / 60;
    seconds = rem_seconds % 60;

    printf("%lld seconds = %d Days, %d Hours, %d Minutes, %d Seconds\n",
        total_seconds, days, hours, minutes, seconds);

    return 0;
}
```

```
Last login: Thu Sep 3 14:32:05 on console
Restored session: Wed Sep 2 23:43:45 IST 2020
dhruvpandey@DHruV's-MacBook-Air: ~ % nano day18.c
dhruvpandey@DHruV's-MacBook-Air: ~ % clang day18.c -o day18
dhruvpandey@DHruV's-MacBook-Air: ~ % ./day18
Enter time in seconds: 12
12 seconds = 0 Days, 0 Hours, 0 Minutes, 12 Seconds
dhruvpandey@DHruV's-MacBook-Air: ~ % ./day18
Enter time in seconds: 34
34 seconds = 0 Days, 0 Hours, 0 Minutes, 34 Seconds
dhruvpandey@DHruV's-MacBook-Air: ~ % ./day18
Enter time in seconds: 7
7 seconds = 0 Days, 0 Hours, 0 Minutes, 7 Seconds
dhruvpandey@DHruV's-MacBook-Air: ~ % ./day18
Enter time in seconds: 99
99 seconds = 0 Days, 0 Hours, 1 Minutes, 39 Seconds
dhruvpandey@DHruV's-MacBook-Air: ~ % nano day18.c
dhruvpandey@DHruV's-MacBook-Air: ~ % clang day18.c -o day18
dhruvpandey@DHruV's-MacBook-Air: ~ % ./day18
Enter time in seconds: 45
45 seconds = 0 Days, 0 Hours, 0 Minutes, 45 Seconds
dhruvpandey@DHruV's-MacBook-Air: ~ % ./day18
Enter time in seconds: 3758
3758 seconds = 0 Days, 1 Hours, 2 Minutes, 38 Seconds
dhruvpandey@DHruV's-MacBook-Air: ~ % ./day18
Enter time in seconds: 98861
98861 seconds = 1 Days, 1 Hours, 1 Minutes, 1 Seconds
dhruvpandey@DHruV's-MacBook-Air: ~ %
```

PART 2: LEAP YEAR CHECKER (COMPACT & LOGICAL OPTIMIZED)

2.1 Algorithm

1. Start

2. **Input:** Read year.

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3. **Condition Check:** If $(\text{year} \% 4 == 0 \text{ AND } \text{year} \% 100 != 0)$
OR $(\text{year} \% 400 == 0)$, flag as Leap Year.
4. **Else:** Flag as Not a Leap Year.
5. **Output:** Print validation result.

6. Stop

2.2 Pseudocode

text

BEGIN

NUMBER year

OUTPUT "Enter a year: "

INPUT year

IF $((\text{year} \% 4 == 0 \text{ AND } \text{year} \% 100 != 0) \text{ OR } (\text{year} \% 400 == 0))$ THEN

OUTPUT year, " is a leap year."

ELSE

OUTPUT year, " is not a leap year."

ENDIF

END

Use code with caution.

2.3 Flowchart Text Representation

text

[Start]

|

▼

[Input year]

|

▼

/-----\

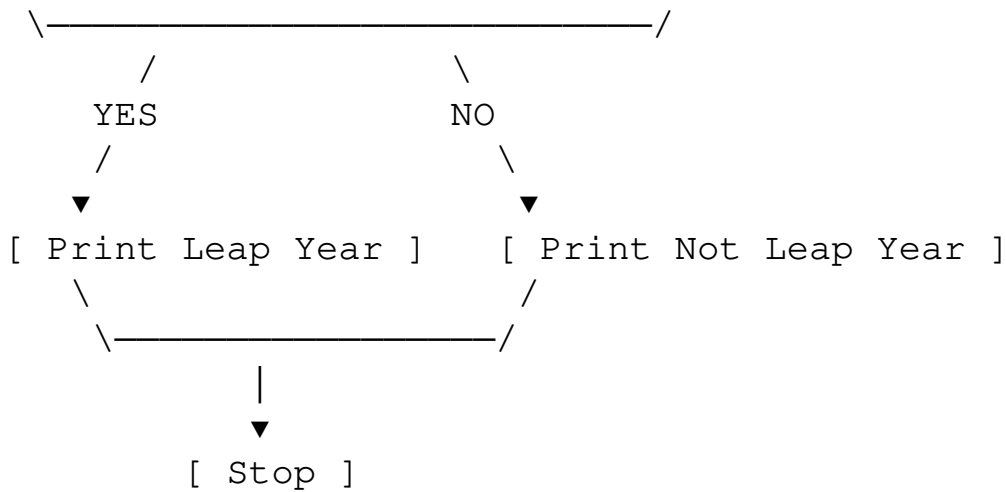
< $(\text{yr}\%4==0 \ \&\& \ \text{yr}\%100!=0)$ >

< || $(\text{yr}\%400==0)?$ >

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Use code with caution.

2.4 Test Case Table

Test Case ID	Input (Year)	Expected Output	Status
TC2_01	2024	2024 is a leap year.	Pass
TC2_02	1900	1900 is not a leap year.	Pass
TC2_03	2000	2000 is a leap year.	Pass
TC2_04	2023	2023 is not a leap year.	Pass

2.5 Source Code (`leap_year.c`)

c

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The image shows a code editor window titled "LW PICO 5.09" with a file named "day10.1.c". The code is a C program that checks if a year is a leap year. It includes `<stdio.h>` and defines a `main` function. The program prompts the user to enter a year, checks if it's a valid year (greater than 0), and then uses a consolidated optimal condition to determine if it's a leap year. The condition is `((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))`. If it's a leap year, it prints "is a leap year.", otherwise, it prints "is not a leap year.". The program returns 0.

```
#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    if (scanf("%d", &year) != 1 || year < 0) {
        printf("Invalid input. Please enter a valid year.\n");
        return 1;
    }

    // Consolidated optimal condition check
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("%d is a leap year.\n", year);
    } else {
        printf("%d is not a leap year.\n", year);
    }

    return 0;
}
```

Below the code editor is a terminal window showing the compilation and execution of the program. The user runs `clang day10.1.c -o day10.1` to compile the program. Then, they run `./day10.1` to execute it. The terminal shows the program's output for several test cases: 2024 is a leap year, 1900 is not a leap year, 2000 is a leap year, and 2023 is not a leap year.

```
dhruvpandey@DHruVs-MacBook-Air ~ % clang day10.1.c -o day10.1
dhruvpandey@DHruVs-MacBook-Air ~ % ./day10.1
Enter a year: 2024
2024 is a leap year.
dhruvpandey@DHruVs-MacBook-Air ~ % ./day10.1
Enter a year: 1900
1900 is not a leap year.
dhruvpandey@DHruVs-MacBook-Air ~ % ./day10.1
Enter a year: 2000
2000 is a leap year.
dhruvpandey@DHruVs-MacBook-Air ~ % ./day10.1
Enter a year: 2023
2023 is not a leap year.
dhruvpandey@DHruVs-MacBook-Air ~ %
```

PART 3: COMPILATION & EXECUTION INSTRUCTIONS

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To complete the execution step required for Day-10, students can run the following terminal sequences:

bash

```
# Compile the programs
gcc time_converter.c -o time_converter
gcc leap_year.c -o leap_year
# Execute the programs
./time_converter
./leap_year
```